Image Enhancement using linear filters for Rectal Cancer Diagnosis using MR Images

Sonal Patreena D'Almeida
Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education
Manipal, India
sonaldalmeida96@gmail.com

Dr. Surekha Kamath
Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education
Manipal, India
surekha.kamath@manipal.edu

Dr. Rajagopal K.V
Department of Radiodiagnosis and Imaging, Kasturba Medical College, Manipal Academy of Higher Education,
Manipal, India
manjunath.kn@manipal.edu

Dr. Manjunath K.N
Department of Computer Science and Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education,
Manipal, India
rajagopal.kv@manipal.edu

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***Abstract*—Computer-aided diagnosis (CAD) is an efficient tool in analyzing medical images for the diagnosis and staging of rectal cancer. Magnetic Resonance Imaging (MRI) plays a crucial role in the detection and evaluation of rectal cancer due to its high-resolution imaging capabilities. This paper discusses the pre-processing methods for MR images of rectal cancer, which facilitate further image analysis for diagnosing the stage of cancer. Pre-processing is a critical step for MR images, aiding in subsequent steps of segmentation and classification. This paper investigates the impact of widely used pre-processing techniques. Initially, pelvic DICOM images are provided as input. By combining three filters—mean, median, and adaptive median filters—the noise is removed from the image. Image enhancement techniques such as Adaptive Histogram Equalization (AHE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) are used to enhance the quality of the image. Statistical parameters like Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Structural Similarity Index Measure (SSIM) are used to evaluate performance. Results indicate that CLAHE provides superior outcomes.**

Keywords— Computer-aided diagnosis, Magnetic Resonance Imaging, AHE, CLAHE, PSNR, MSE, RMSE, SSIM

I. INTRODUCTION

MRI plays an important role in medical diagnosis and research. Processing is often required before raw MRI data can be used effectively. Preprocessing aims to improve image quality, accuracy, and performance of machine learning models. The MRI is stored in DICOM format, which stores image data and metadata. Typically, preprocessing begins with loading DICOM files and converting them to a more manageable format for graphics libraries, such as NIfTI or even converting them to a simple format [1]. Denoising techniques such as filtering or nonlocal instrumental filtering help to remove this noise while preserving the underlying features of the image. Normalization techniques such as histogram equalization or intensity scaling aim to bring these intensities into the standard range across the data set, making image analysis more likely as MRI scans can be affected by artifacts such as noise, intensity heterogeneity (brightness variation) and distortion. Pre-processing techniques can reduce noise, correct fluctuations in intensity, and improve image uniformity, providing clear anatomical information. Many medical image analysis tasks, like tumor segmentation or organ delineation, rely on algorithms that automatically identify specific structures. Preprocessing helps prepare the images for these algorithms by increasing the contrast resulting in more accurate segmentation results [2].

II. LITERATURE REVIEW

Authors proposed several kinds of pre-processing methods for image enhancement and noise reduction [3]. High pass filters are used to sharpen images that have been contrast-enhanced. Several types of noise, represented as Poisson, Gaussian, salt-and-pepper, and speckle noise are taken into consideration. The adaptive median filters, mean and median filters are used to remove noise. MSE and PSNR are used to determine the ideal filter.

Authors developed a new method that adds features to the current median filter [4]. The three image filtering algorithms like Median, Max filter and Gaussian filter, Min filter and Arithmetic Mean filter were used to examine the experimental results. Statistical metrics such as SNR, PSNR and root mean square error (RMSE) were employed.

The reproducibility of quantitative imaging features or radiomics, derived from apparent diffusion coefficient (ADC) maps of patients with rectal cancer is examined [5]. The study involved 56 patients from two different clinics, which focused on the impact of noise present in ADC images, manual tumor delineation, pixel size resampling, and intensity discretization on various features. Results suggest that intensity distribution histogram-derived features are less sensitive to manual delineation and image perturbations.

Authors proposed a CNN model which is equipped with a multi-layer structure designed to eliminate salt and pepper noise [6]. The structure incorporates batch normalization, padding, and a rectified linear unit. In a transformative step, all pixels with values of 255 are substituted with 0, thereby transforming bipolar salt and pepper noise into unipolar noise.

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This alteration accelerates the training process and enhances denoising performance. The denoiser proves beneficial for images plagued by a substantial number of interference pixels, mitigating the risk of misjudgment and minimizing errors.

In paper [7] authors proposed various contrast enhancement techniques, such as knee function and gamma correction based on Singular Value Decomposition with Discrete Wavelet Transform (SVD-DWT) method (KGC-DWT-SVD), Global Histogram Equalization (GHE) and Adaptive Gamma Correction (AGC). These techniques were used on MR images of spinal cord and brain in multiple sclerosis patients. Entropy (H), PSNR, MSE, FSIM, SSIM parameters were assessed.

Authors presented a pre-processing method for brain tumor detection that resamples images after applying translation and geometric correction, which causes pixels to be redistributed according to their spatial displacements [8]. Three sampling algorithms—Bilinear Interpolation approach, Nearest Neighbor technique and the Cubic Convolution technique are used to obtain resampling and Enhancement of Gray Scale Contrast, Markov Random Field is used for noise reduction.In paper [9] a method for pre-processing the brain tumor image using techniques for image enhancement and noise reduction is suggested, which improves the image quality for edge detection and precise tumor positioning. Gaussian high pass filter is used to improve input image, and filters are used to extract noise from the slice in order to sharpen the image.

In paper [10], the authors developed a method for enhancing medical images using AHE and CLAHE. Statistical parameters like MSE, PSNR, Normalized Cross-Correlation, Average Difference, SSIM, Max Difference were used to evaluate the performance. CLAHE reduced noise and achieved better contrast.

Authors explored image enhancement techniques to compare three methods: HE, AHE, and CLAHE [11]. The study employed image quality metrics like MSE, PSNR, SNR to evaluate the enhanced images. Based on these metrics, CLAHE had better results. Authors developed RACLAHE [12]. Unlike existing methods like CLAHE that enhanced the entire image, RACLAHE focused on the prostate region. It achieved this by first identifying the prostate using a U-Net like structure and reduced the image size to focus on relevant features. Patch-wise CLAHE approach was applied on RACLAHE for the prostate region, analyzing pixel intensities within small patches and calculating new intensity values based on the patch's histogram.

III. METHODOLOGY

Preprocessing plays an important role in optimizing pelvic MR images for clinical examination, diagnosis, and treatment planning. Denoising and image enhancement are important preprocessing steps because they significantly improve the quality and clarity of images. It reduces noise that can obscure important features, and it enhances detail, resulting in greater contrast and more visually appealing images. This preprocessing is essential for accurate feature extraction, segmentation, and analysis, and results in better performance in machine learning models.

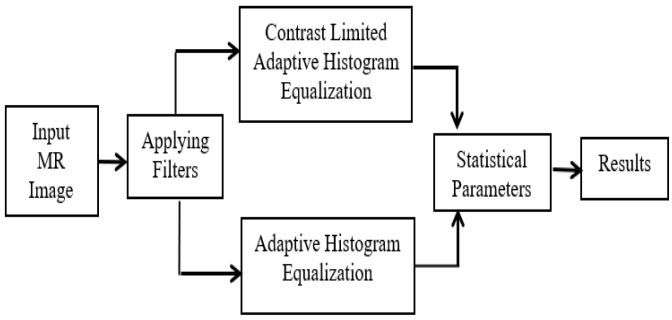

Figure 1: Workflow diagram of proposed model

In the proposed method, pelvic MR images is provided as input. The proposed method was developed by applying a combination of mean, median, and adaptive mean filters, which aimed to improve the overall image quality. Preprocessing is performed in order to lower noise, enhance image quality, and get input ready for further process optimization. The resulting image is then subjected to CLAHE and AHE as shown in the fig. 1. Statistical parameters such as PSNR, MSE, RMSE, and SSIM are computed to quantitatively evaluate the effectiveness of the enhancement process. This comprehensive approach aims to optimize contrast and improve the visibility of anatomical details in pelvic MRI image.

A. Proposed Method:

The proposed method involves combination of adaptive median, mean, and median filters. It can provide comprehensive noise reduction. Each filter addresses different types of noise and smoothing is necessary. The adaptive medium filter deals with high levels of noise, the mean filter smooths the image, and the median filter ensures that fast noise is reduced. These filters have been used primarily for their effectiveness at noise levels in a digital image. While other filters such as linear filters or frequency domain filters provide exceptional noise reduction capabilities, they often require more complicated implementation and may be less effective in dealing with noise conditions.

i. Input DICOM Image

The process begins with the input of a DICOM image, which contains MRI images and other medical information.

ii. Converting to Grayscale

System checks if the DICOM image is in grayscale format. If it's grayscale, then filtering steps. If it is not grayscale, convert the DICOM image to grayscale. This step ensures perfect performance with DICOM models.

iii. Applying Adaptive and Traditional Filters

a) Adaptive mean filter: Apply an adaptive mean filter to a grayscale DICOM image. Locally adjusts filter size, preserves edge detail and reduces noise in texture areas.
b) Median Filter: Apply a median filter to a grayscale DICOM image. It replaces the value of each pixel with the mean value of its

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neighbor, and is effective against salt and pepper noise.

c) Mean Filter: Apply a mean filter to a grayscale DICOM image. It calculates the average value of neighboring pixels, smoothing out the noise and making it uniform.

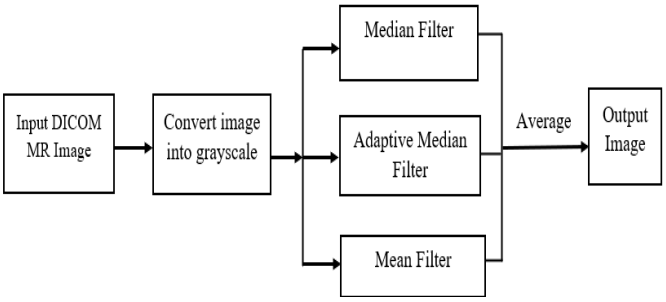

Figure 2: Workflow diagram of combined filters

iv. Combine Filter Outputs

a) Combine the results of the adaptive median, median, and mean filters as shown in fig.2.
b) Averaging ensures a balanced approach to noise reduction and image enhancement:
c) Adaptive median retains sharp edges in complex textures.
d) Median filter tackles impulse noise effectively.
e) Mean filter provides overall smoothing, beneficial for noise reduction in uniform regions.

computing the mean of three filters as shown in eq.1- Mean (F_M), Median (F_{Med}), and Adaptive Median (F_{AdMed}) using,

$$I_f = \frac{F_M(I) + F_{Med}(I) + F_{AdMed}(I)}{3} \ldots\ldots\ldots\ldots (1)$$

where:

- $F_M(I)$ is the Mean Filtered image,
- $F_{Med}(I)$ is the Median Filtered image,
- $F_{AdMed}(I)$ is the Adaptive Median Filtered image,
- I_f is the final noise-reduced image after averaging.

v. Applying CLAHE and AHE for Enhancement

Histogram equalization is a technique for contrast enhancement. It is an image processing technique used to increase image contrast by redistributing pixel intensity values in a histogram. It is a histogram which is then normalized by calculating the cumulative distribution function. Using the normalized CDF old energy values can be mapped to new ones. AHE and CLAHE are the advanced variations.

AHE is an extension of the histogram equalizer that works in small areas of the image, making it more suitable for images of varying brightness. It increases image contrast by redistributing intensity values that maximize overall contrast. AHE divides the image into smaller regions and performs histogram equalization independently in each region.

CLAHE is an advanced image enhancement technique that can be used to improve the contrast of images while avoiding excessive noise amplification. It divides the image into tiles and makes the histogram on each tile the same. This allows for increased local contrast, which is particularly useful in images with uneven illumination or with different levels of contrast in different regions CLAHE adds a filter to prevent harsh intensity limit, so that noise does not cause excessive noise levels. Furthermore, by adjusting the clip boundary parameters, the CLAHE can be optimized to achieve the desired improvement, allowing flexibility in different imaging conditions. After noise removal, contrast enhancement is applied for CLAHE (E_{CLAHE}) and AHE (E_{AHE}) as shown in eq.2 and eq.3-

$$I_{CLAHE} = E_{CLAHE}(I_f) \ldots\ldots\ldots\ldots (2)$$

$$I_{AHE} = E_{AHE}(I_f) \ldots\ldots\ldots\ldots\ldots (3)$$

where:

- $E_{CLAHE}(I_f)$ is the CLAHE-enhanced image.
- $E_{AHE}(I_f)$is the AHE-enhanced image.

vi. Output Preprocessed DICOM Image

The final output of the process is a preprocessed DICOM image that represents the end of the filtering step for DICOM format and medical image analysis needs Clarity, noise reduction and prepares the image for semantic interpretation disease or other computer screening as shown in fig.2.

IV. RESULT AND DISCUSSION

MR Images of the rectum were obtained from the datasets acquired at KMC, Manipal. The preprocessing method applied to these images ensures clarity, free from artifacts, and without any loss of vital information. The image preprocessing was performed on 2D slices of DICOM images, resulting in high-quality images suitable for further analysis.

This paper presents an analysis of a proposed method that combines adaptive median filtering, mean filtering, and median filtering to effectively address different types of noise in digital images. This specific filter was chosen due to its ability to effectively handle different noise levels. Through the combination of these filters, a balanced approach to noise reduction is achieved, efficiently reducing noise while maintaining significant information in the image.

The mean filter uses a specific kernel to average the pixel values, smoothing the images. This reduces noise but obscures features and edges. It smooth's the image by reducing random noise as shown in fig.3.

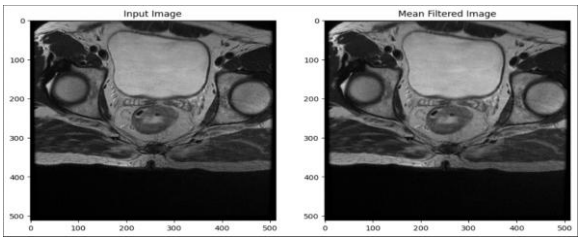

Figure 3: Results of mean filtered image.

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By substituting the center of each neighboring pixel at a given kernel for each pixel value, the median filter smoothes the images. By efficiently reducing noise, edges are preserved as shown in fig.4.

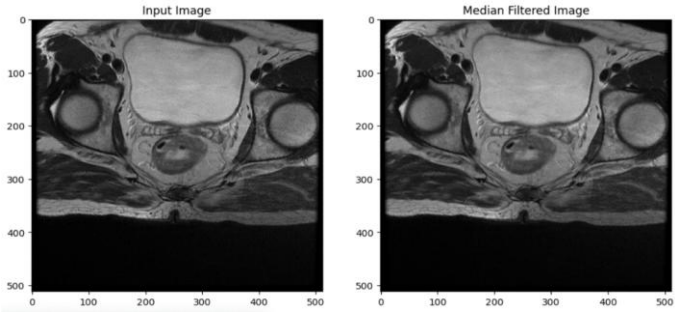

Figure 4: Results of median filtered image

The adaptive mean filter adjusts the filtering window size based on local noise characteristics. It replaces each pixel value with the mean value of the adaptive window, which is propagated when noise is detected. This method effectively removes various types of noise, including salt and pepper noise, and preserves edges and finer details better than standard median filters as shown in fig.5.

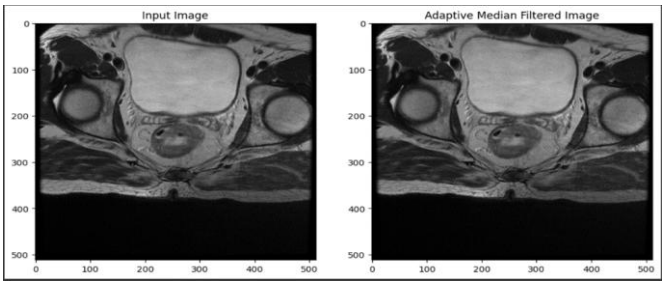

Figure 5: Results of adaptive median filtered image

The proposed method integrates Mean, Median, and Adaptive Median filters for effective image noise reduction. This process is further enhanced using image enhancement techniques such as AHE and CLAHE, as illustrated in Fig. 6.

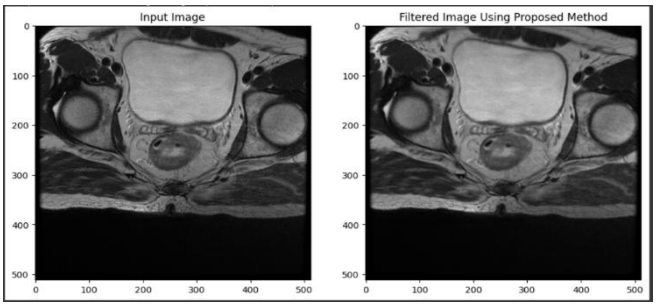

Figure 6: Results of Proposed method

Results obtained from the mean, median, adaptive median filter and proposed method are shown in table 1. It is observed that proposed method had better results.

TABLE 1: COMPARISON BETWEEN THE FILTERS AND PROPOSED METHOD

Image Type	*Methods*	*PSNR*	*SSIM*	*MSE*	*RMSE*
MR Rectal Images	Mean filter	59.4745	0.9985	4.5475	0.0021
	Median Filter	52.1857	0.9978	6.0870	0.0024
	Adaptive Median Filter	47.9250	0.9947	1.6245	0.0040
	Proposed Method	57.2505	0.9974	7.5859	0.0027

The results presented in table 2, demonstrate enhancement techniques refine overall image quality by optimizing contrast and emphasizing key features.

TABLE 2: STATISTICAL RESULTS OBTAINED FROM IMAGE ENHANCEMENT TECHNIQUES

Image Enhancement Techniques	*PSNR*	*SSIM*	*RMSE*
CLAHE	58.23	0.9956	0.0021
AHE	53.21	0.9921	0.0027

V. CONCLUSION

To improve diagnostic accuracy, modern image processing and machine learning techniques are used in computer-aided diagnosis (CAD_x) for rectal cancer. CAD_x is a promising tool for designed treatment plans and early rectal cancer detection. In order to eliminate noise from medical images, the study proposed an approach that combines adaptive median, mean, and median filters. The results have demonstrated that noise could be successfully eliminated from the medical images while maintaining their structural integrity. To improve the image quality further, image enhancement techniques like AHE and CLAHE were applied. Pelvic MR Images from KMC, Manipal were used for evaluating the filtering methods. To assess the image quality, metrics like PSNR, MSE, RMSE, and SSIM were employed. The outcomes showed that, in comparison to other techniques, CLAHE achieved better enhancement. By increasing the accuracy of MR images, this method will help and strengthen diagnosis. The study has certain limitations, the filters may smooth out fine anatomical details, potentially affecting tumor boundary preservation. Adaptive Median filtering increases computational complexity, making it less efficient for large datasets. CLAHE and AHE can sometimes introduce artificial noise or over-enhance certain regions, leading to intensity inhomogeneity. In the future, noise reduction in rectal cancer imaging can be enhanced by combining several filtering systems.

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